The MeV-region capture budget for X17 production at n_TOF EAR2

What 5×10^8 fully simulated neutrons (1 meV–100 MeV) say about moving the analysis window

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Internal note — informal. Drafted by Claude (Anthropic AI assistant) under D. Neff's direction; simulation, numbers and text reviewed by D. Neff.

Update 2026-07-27 (nose-first + final geometry). This is the 12 June draft. **Production is unchanged** (∼32 X17/day over the MeV region is gas physics). **Superseded:** the in-gate detector-background counts below are pre-final-geometry (STEP LS vessels, 2 cm plastics, mid-July) — in particular the Al-wall term scales ×0.577 nose-first (∼255→∼147 captures/pulse). The recorded-yield / MM-double acceptance / significance numbers now live in `angular_note.tex` (nose-first: acceptance 27.8/27.4%, Asimov $Z = 3.5/6.4\sigma$ July/LS3). See `CAMPAIGN_STATUS.md`.

Part I — Summary

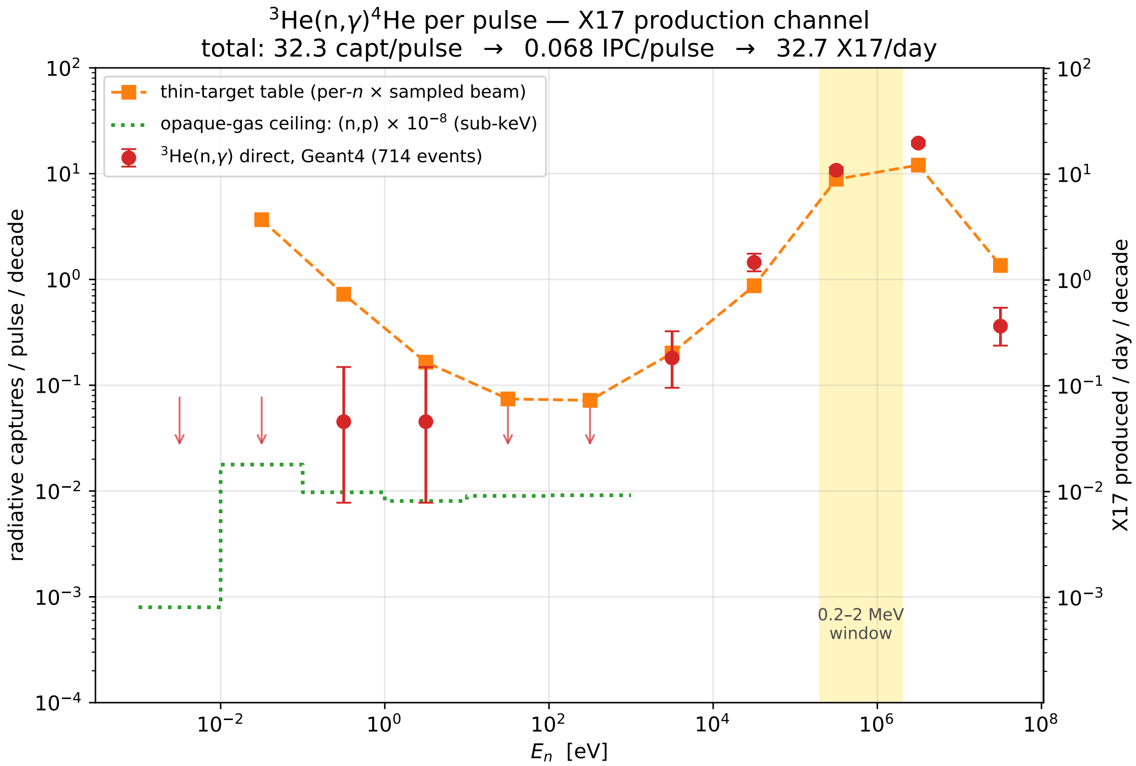
The question

The thermal note (11 June) showed that the sub-keV X17 measurement is dead: below 1 keV the ^{3}He gas is opaque to (n,p)t, radiative captures are capped at $\sigma_{n\gamma}/\sigma_{np} \approx 10^{-8}$ per beam neutron, and the yield is ∼0.1 X17/day — a factor 50–100 below the rate table. The rescue plan was always written in the same table: above ∼100 keV the gas is thin, the thin-target arithmetic is honest there, and those rows carry ∼98% of the physical rate. **This note asks whether the full simulation confirms that promise** — and delivers the requested curve of expected X17/day as a function of neutron energy, before detector acceptance.

What we did

Run B-full fired 5×10^8 neutrons, sampled from the evaluated EAR2 flux over the *entire* range 1 meV–100 MeV with the energy-dependent transverse beam profile, at the full STEP-derived capsule and detector geometry (G4NDL high-precision neutron data). Unlike the sub-keV run, here the radiative channel is measured *directly*: **714 ^{3}He(n,γ)^{4}He events** were recorded — 95% of them above 100 keV, median $E_n = 1.3$ MeV. No branching-ratio extrapolation is needed; the per-decade rate below is simply the direct count. Normalisation: the sampled flux histogram integrates to 2.26×10^7 n/pulse over the full range (its sub-keV part reproduces the known 7.31×10^6 anchor), and 1.929×10^4 pulses/day.

The answer



The direct counts (red) tell a complete story in one plot. Below 1 keV they sit on the opaque-gas ceiling (green), ×8–80 under the table — the thermal-note result, independently reproduced here (2 events → 0.09 capt/pulse, vs. (0.5–1.1) × 10^{-1} from the dedicated 10^9-event sub-keV run). Above ∼10 keV the data *meet and slightly exceed* the thin-target table (orange): the gas has turned thin and the table rows are valid, as claimed. Integrated, with the 2.5% X17/IPC branching applied *per IPC pair*:

	per pulse		per day	
E_n = 1 meV–100 MeV	0.2–2 MeV window	full range	0.2–2 MeV window	full range
radiative captures	22.2 ± 1.0	32.3 ± 1.2	4.3×10^5	6.2×10^5
IPC pairs ($\times 2.1 \times 10^{-3}$)	4.7×10^{-2}	6.8×10^{-2}	899	1309
X17 (2.5% of IPC)	1.2×10^{-3}	1.7×10^{-3}	**22.5 ± 1.0**	**32.7 ± 1.2**

The bottom line: ∼33 X17 produced per day over the full range, of which ∼22.5/day land in the single 0.2–2 MeV window — before acceptance, trigger and analysis efficiency. That is a factor ∼300 above the sub-keV yield (0.06–0.11/day), and it agrees with the commented-out 0.2–2 MeV row of `results_3He` (24.9 X17/day) at the 10% level. Where the table is valid, the full simulation *confirms* it; per incident neutron Geant4 actually finds ×1.2–1.7 *more* captures than the table's 4-cm-sphere model, consistent with the real capsule's longer on-axis gas column and scattering re-entry. A 30-day run produces ∼700 X17 decays in the window. **The measurement moves from impossible to clearly feasible — the remaining questions are acceptance and backgrounds, not production.**

Quoted errors are statistical only. The dominant systematics are inherited assumptions: the IPC coefficient (2.1×10^{-3}/capture) and the X17/IPC ratio (2.5%), both still awaiting provenance from Alberto, and the flux normalisation (the Ph3 evaluated flux used here is 31% below the table's 3.29×10^7 n/pulse; per-neutron physics is unaffected).

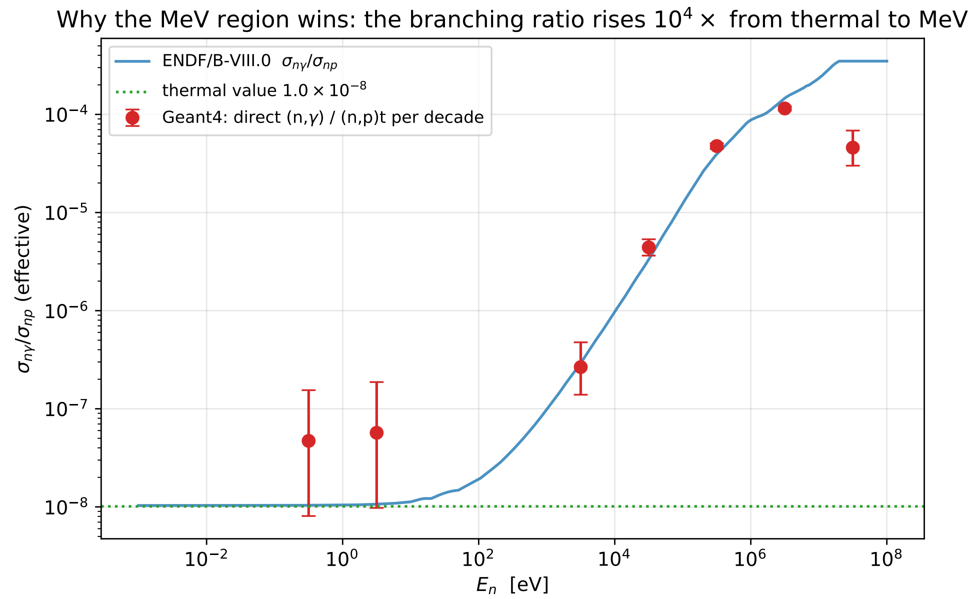
What's next

The at-rest pair sample (`pairs_v2_step_target`, 10^7 events) gives the detector-response machinery, but MeV captures form $^4\mathrm{He}^*$ at $E^* \approx 20.58\,\mathrm{MeV} + \frac{3}{4}E_n$ with a CM boost — the generator needs E_n-dependent kinematics before final acceptance numbers. In time, the window sits 1.0–3.2 μs after the γ flash at EAR2 ($\sim$19.5 m), sharing the gate with 3×10^5 (n,p)t and 9×10^4 scintillator H-captures per pulse (Sec. 5–6).

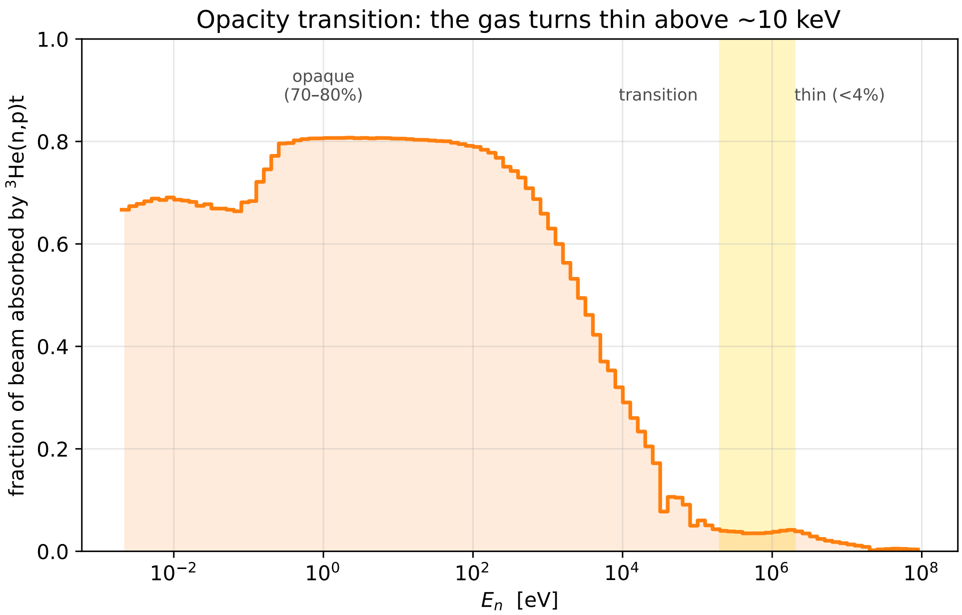
Part II — The full story

1 Why the rate comes back at MeV energies

The sub-keV failure was a competition problem: every neutron is absorbed, so radiative capture can never beat $\sigma_{n\gamma}/\sigma_{np} \approx 10^{-8}$. But that ratio is not a constant of nature — it is a thermal-energy accident. σ_{np} falls as $1/v$ while $\sigma_{n\gamma}$ picks up direct-capture strength, so the branching ratio climbs four orders of magnitude between thermal and MeV:



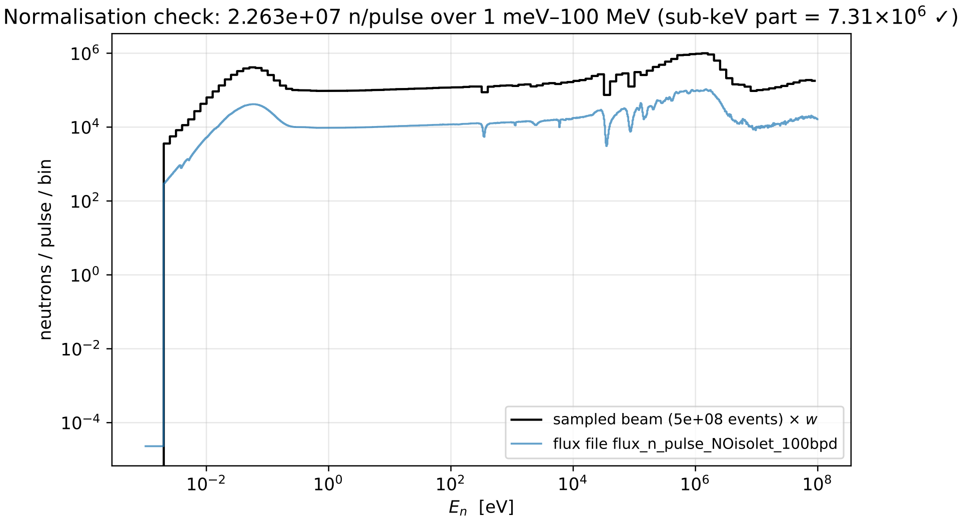
The Geant4 per-decade ratios (direct (n,γ) over (n,p)t, red) track the ENDF/B-VIII.0 evaluation across six decades — a non-trivial closure test of the simulation's neutron physics. (The 10–100 MeV point falls low: G4NDL data end at 20 MeV and other channels open; with 8 events and 0.4 X17/day it is irrelevant to the budget.) Two things then happen simultaneously above ∼100 keV: the branching improves by 10^4, *and* the gas stops eating the beam:



Below 1 keV the gas absorbs 67–81% of the beam (opaque); by 100 keV it absorbs <4% (thin). This is the transition the thin-target table silently assumes — and above it, the table is right.

2 The run and its normalisation

Run B-full: 5×10^8 events, neutron-beam mode, energies sampled from `flux_n_pulse_NOisolet_100bpd` (`data/fluxEAR2-Ph3_in_different_units.root`) over 1 meV–100 MeV, transverse positions from the Lambda2D profile, STEP capsule geometry. All 100 output files validated (tree readable, entry counts exact). Each simulated neutron carries weight $w = 2.263\times10^7/5\times10^8 = 4.53\times10^{-2}$ per pulse.



Two anchor checks: the sampled spectrum reproduces the flux file bin-by-bin (the generator samples that exact histogram), and the sub-keV integral is 7.312×10^6 n/pulse — the anchor used throughout the thermal analysis. Note this evaluated flux is 31% below the 3.29×10^7 n/pulse assumed in `results_3He`; comparisons with the table are therefore made *per incident neutron*, and per-pulse rates quoted here are the ones consistent with the Ph3 flux.

3 The capture budget, decade by decade

The requested deliverable — expected X17/day versus neutron energy. (A pleasant numerical accident: with $\alpha_{\mathrm{IPC}}\cdot\mathrm{BR}_{\mathrm{X17}}\cdot N_{\mathrm{pulses/day}} = 2.1\times10^{-3}\times0.025\times1.929\times10^4 = 1.01$, captures/pulse and X17/day are numerically equal, so the table reads either way.)

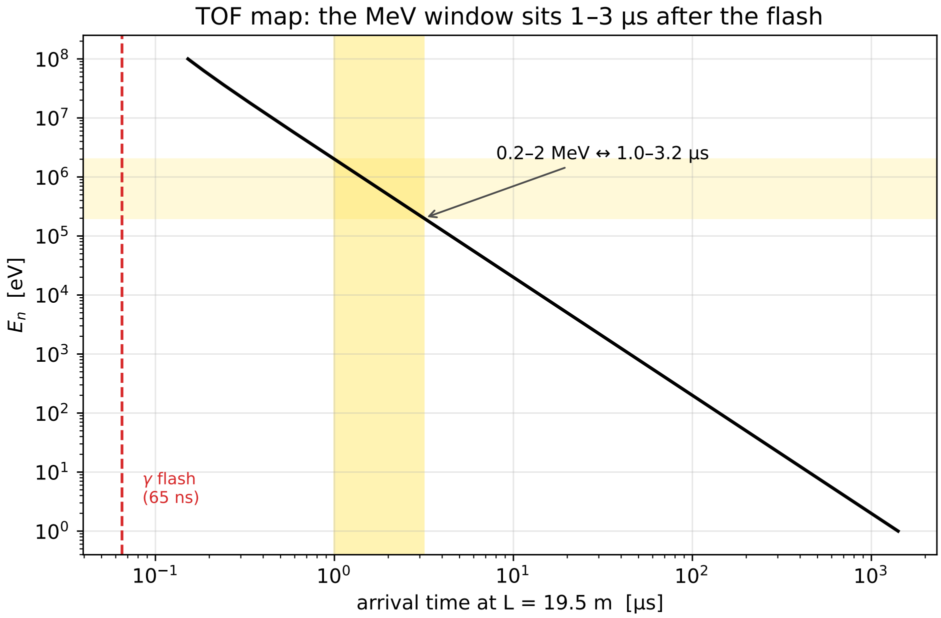
decade	beam/pulse	(n,p)/beam	$N_{n\gamma}$	capt/pulse	X17/day	G4/table
1 meV–10 meV	1.15×10^5	0.69	0	<0.08	<0.08	—
10 meV–0.1 eV	2.61×10^6	0.67	0	<0.08	<0.08	opaque
0.1–1 eV	1.27×10^6	0.76	1	0.05	0.05	0.06
1–10 eV	9.85×10^5	0.81	1	0.05	0.05	0.27
10–100 eV	1.11×10^6	0.80	0	<0.08	<0.08	opaque
0.1–1 keV	1.22×10^6	0.74	0	<0.08	<0.08	opaque
1–10 keV	1.45×10^6	0.47	4	0.18	0.18	0.90
10–100 keV	2.00×10^6	0.16	32	1.45	1.47	1.66
0.1–1 MeV	5.84×10^6	0.039	238	10.8	10.9	1.22
1–10 MeV	4.59×10^6	0.037	430	19.5	19.7	1.61
10–100 MeV	1.43×10^6	0.006	8	0.36	0.37	0.27
total	2.26×10^7		714	32.3	32.7	

($N_{n\gamma}$ = direct counts; upper limits are 68% CL for empty decades; "G4/table" is the ratio of per-neutron capture probabilities.) Three regimes are visible. **Opaque** (below ∼1 keV): the simulation sits far under the table, on the branching-ratio ceiling — 2 events here vs. the table's would-be ∼290. **Transition** (1–100 keV): absorption falls from 47% to 16%, the table becomes honest (G4/table 0.9 → 1.7). **Thin** (above 100 keV): 95% of all production, table confirmed.

4 Checking the table where it claims validity

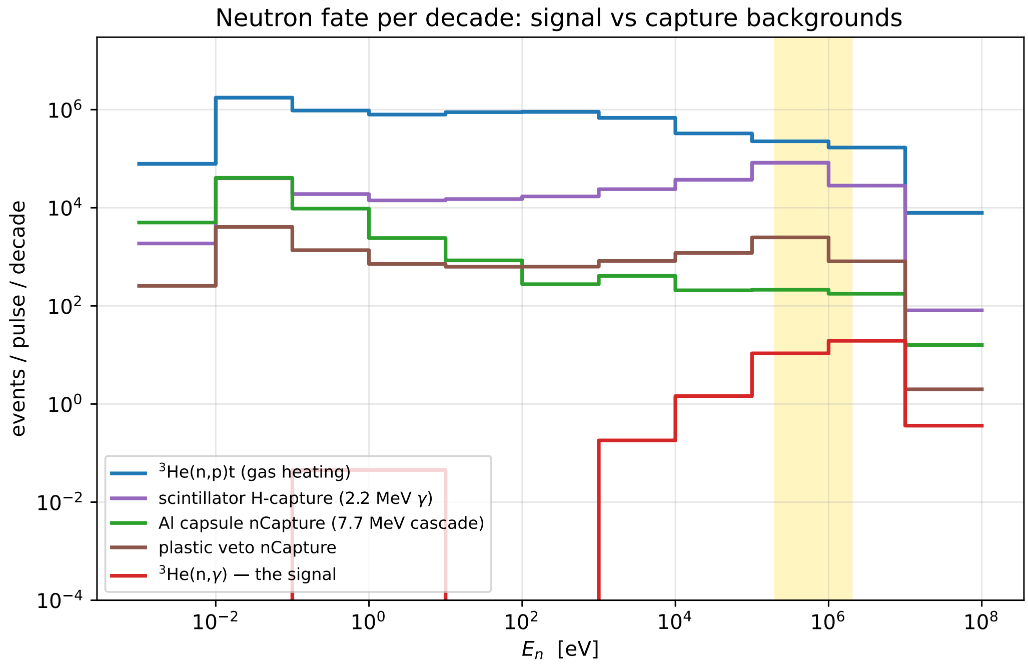
In the two dominant decades the per-neutron agreement is $\times 1.22$ (0.1–1 MeV, 238 events) and $\times 1.61$ (1–10 MeV, 430 events), Geant4 *above* the table. The excess has a mundane reading: the table models a 4 cm ^{3}He sphere (0.013 mm equivalent thickness) in isolation, while the real capsule presents up to 60 mm of gas on axis plus a chance for neutrons scattered in the surrounding apparatus to re-enter. The sign and size are what one expects, and for planning purposes the conservative number is the table's own. The 0.2–2 MeV window integral — 490 direct events, 22.2 ± 1.0 capt/pulse — lands within 10% of the commented table row (24.6), the row that originally motivated the move.

5 The window in time



At EAR2's ∼19.5 m, 0.2–2 MeV neutrons arrive 1.0–3.2 μs after production — a 2.2 μs gate, 15–50 flash transit times after the γ flash (65 ns). Whether the detectors have recovered from the flash by 1 μs is an *experimental* question this simulation cannot answer (the flash itself is not modelled); it is the single most important measurement to make with the real setup. Within the gate, per pulse, the simulation puts: 3.0×10^5 (n,p)t events in the gas (764 keV of charged heat each — the dominant in-window energy deposition), 8.8×10^4 H-captures in the liquid scintillator (2.2 MeV γ, born *inside* the calorimeter), 2.6×10^3 plastic-veto captures and ∼255 Al-wall captures (7.7 MeV cascades; nose-first ∼147, $\times 0.577$). The signal: 22 IPC-capable captures per pulse, of which 4.7×10^{-2} produce a pair. Pile-up and trigger design at μs timescales, not raw rate, will set the sensitivity.

6 Backgrounds across the spectrum



The full-range hierarchy per pulse: 67% of the beam escapes without a terminal interaction (mostly fast neutrons punching through the thin gas), 30% undergoes (n,p)t (concentrated sub-keV, where the late-arriving neutrons deposit their heat tens of μs to ms after the window of interest — a slow-tail rate issue, not an in-window one), and the leading capture backgrounds are the liquid-scintillator H-captures (2.8×10^5/pulse) and Al-capsule captures (6.0×10^4/pulse) — the same two channels flagged in the thermal note, now quantified over the full spectrum. Run C (biased wall-γ source) remains the right tool to turn these counts into calorimeter spectra.

7 What is still assumed

1. **IPC coefficient** 2.1×10^{-3}/capture and **X17/IPC = 2.5%**: flat values from `results_3He`, provenance (multipolarity, E^* dependence) still to be confirmed with Alberto. Every per-day X17 number scales linearly with both.

2. **Kinematics**: MeV captures form $^4\mathrm{He}^*$ at $E^* = 20.58\,\mathrm{MeV} + \frac{3}{4}E_n$ (+ CM boost): at 2 MeV that is 1.5 MeV of extra excitation and a boosted decay frame — opening angle, energy sharing and TOF of the pair all shift. The existing at-rest pair sample is the right machinery test but the wrong final answer; the generator extension is the gating item for run A.

3. **Acceptance**: nothing here includes the detector. The next step is `analyze_pairs.py` on `pairs_v2_step_target` (10^7 events, current STEP geometry) for the baseline response matrix.

4. **Above 20 MeV**: G4NDL ends; that decade is model-dependent and contributes 1% of the budget.

5. **Gamma-flash recovery**: not simulated; must come from beam data.

8 Where this leaves us

The energy budget argument is settled: **the X17 production rate at EAR2 is ∼33/day, it lives at MeV energies, and the single 0.2–2 MeV window carries ∼22/day** — ×300 the sub-keV yield, in 10%-level agreement with the rate table where the table is valid. The program is now an acceptance-and-backgrounds problem:

1. Pairs acceptance on the at-rest sample (machinery + baseline).
2. Generator extension to E_n-dependent kinematics; re-plan run A.
3. Gamma-flash recovery measurement at EAR2; trigger/pile-up design for a 1–3 μs gate (3×10^5 (n,p) + 10^5 capture γs per pulse in-gate).
4. Run C (wall-γ spectra) and run D (single-particle response) as planned.
5. Alberto: confirm α_{IPC}, X17/IPC, and reconcile the 31% flux-normalisation difference.

Appendix: reproduction

- Scan (lxplus, ∼20 min, 8 workers): `python3 scripts/analyze_mev_captures.py /eos/experiment/nto` `-o mev_captures --workers 8`
- Rates + figures (local or lxplus): `python3 scripts/make_mev_report_figures.py` reads `analysis/mev/mev_captures.npz`, writes `analysis/mev/mev_rates.json` and `docs/report/figs/`.
- Raw scan output: `analysis/mev/mev_captures.npz` (110-bin histograms, terminal-interaction budget, 714 individual (n,γ) energies). Derived rates: `analysis/mev/mev_rates.json`.
- Pitfalls inherited from the thermal campaign: ^{3}He(n,p)t appears as `neutronInelastic` (not `nCapture`); the 2.5% X17 branching is per IPC pair, not per capture; *the thermal* $\sigma_{n\gamma}/\sigma_{np} = 10^{-8}$ *must never be applied outside the sub-keV region* (it is wrong by 10^3–10^4 at MeV — use the direct counts).